SENIOR PROJECT SGN2-2025

Understanding the Effects of Gamification in E-commerce on Motivation, Engagement, and Loyalty

Project Concept

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Project Overview

Abstract

This study investigates the impact of gamification features in e-commerce platforms on consumer behavior; focusing on motivation, user engagement, and loyalty. Given the increasing competition between platforms such as Lazada and Shopee in Thailand, businesses employ gamification mechanisms such as time-limited discounts, point systems membership levels, and product reviews to enhance the user experience. This study aims to determine which gamification features are the most effective method for consumer motivation and engagement, and how these effects may differ across platforms and user groups. The findings are expected to contribute to improving gamification design strategies for e-commerce platforms.

Introduction

E-commerce platforms have become increasingly popular in recent years. However, competition among these platforms are intense as they strive to attract and retain users. To address this, certain companies have adopted the concept of gamification, which applies game-like mechanisms to digital platforms. Examples include: limited-time discounts, point collection systems, membership level upgrades, and product reviews. These features are intended to enhance user motivation, engagement, and loyalty toward the platform.

Despite the growing popularity of gamification, questions regarding how different types of gamification features influence the overall user experience remains to be answered. To be precise, it is not yet fully understood which features most effectively encourage desired consumer behaviors, or how differences between platforms and user demographics affect the impact of gamification in e-commerce.

There is an importance to this study because it explores how gamification elements employed in leading e-commerce platforms such as **Lazada** and **Shopee** influence consumer behavior. The analysis focuses on user engagement, motivation, and brand loyalty. The findings can provide practical insights for developing gamification design strategies that better meet user needs and ultimately improve the effectiveness and competitiveness of e-commerce platforms.

Conceptual Framework

E-commerce platforms employ gamification to enhance user engagement, motivation, and loyalty. However, introducing too many elements can be costly and may overwhelm users. The effectiveness of gamification varies depending on platform characteristics, user experience, and demographic differences. In some cases, gamification features reinforce one another and strengthen engagement, while in other cases they may conflict and reduce effectiveness. Understanding these dynamics is essential for developing strategies that drive long-term loyalty, a key factor in e-commerce profitability.

Gamification Elements and User Experience

- Gamification influences user experience by shaping engagement, intrinsic motivation, and extrinsic motivation, which together contribute to sustained loyalty. The elements can be grouped into three categories:
- Transaction-based: Monetary incentives such as discounts, vouchers, and redeemable points primarily stimulate extrinsic motivation. These features often create short bursts of engagement but may fade once promotions end. In some cases, they can also support intrinsic motivation if users enjoy collecting rewards or participating in time-limited deals.
- Achievement-based: Features like progress bars, challenges, and tiered memberships foster goal-oriented engagement. They provide extrinsic recognition (badges, status levels) while also generating intrinsic satisfaction from progress and accomplishment. Users often remain engaged in pursuit of higher levels and exclusive privileges.
- Social influence-based: Leaderboards, referrals, and review systems encourage participation through competition and community validation. Extrinsic motivation comes from rankings and incentives, while intrinsic motivation arises from belonging and peer recognition. For example, users may write reviews or increase purchases to gain “Top Reviewer” recognition.

These gamification types can interact. Transaction-based rewards can strengthen achievement-based systems, while social influence can enhance both by tying rewards and progress to peer comparison. Effective design requires balancing these elements to maximize engagement while minimizing negative outcomes.

Influence on Loyalty: Mediators and Moderators

- Gamification impacts loyalty primarily through engagement and motivation, which serve as mediators.

- Engagement fosters familiarity and habitual use, encouraging repeat interactions.
- Intrinsic motivation builds emotional attachment and belonging, reinforcing long-term commitment.
- Extrinsic motivation drives continued participation through external rewards, though loyalty may weaken if rewards are perceived as insufficient or temporary.

The strength of these relationships depends on platform design and user demographics, which act as moderators. Younger users may respond more to social features, while frequent shoppers may prefer achievement-based progress systems. By tailoring gamification strategies to user characteristics, platforms can achieve stronger engagement and sustained loyalty.

Requirements

To gather information about users of Shopee and Lazada, a questionnaire was developed that would match elements and criterias that were mentioned above. This data is then translated into table form for easier analytics.

		Count	Column N %
1. Gender :	Female	160	49.5%
	Male	157	48.6%
	Prefer not to say	6	1.9%
2. Age range :	18 - 24	89	27.6%
	25 - 34	128	39.6%
	35 - 44	64	19.8%
	45 - 54	33	10.2%
	Over 55	9	2.8%
3. Marital status :	Single	237	73.4%
	Married	66	20.4%
	Divorced/ Separated/ Widowed	11	3.4%
	Prefer not to say	9	2.8%
4. Current employment status :	Student	113	35.0%
	Corporate employee	99	30.7%
	Government employee	51	15.8%
	Educator	21	6.5%
	Self-employed	29	9.0%
	Unemployed/ retired	5	1.5%
	Other	5	1.5%
5. Average monthly income :	0 - 15,000 THB	100	31.0%
	15,001 - 30,000 THB	61	18.9%
	30,001 - 50,000 THB	79	24.5%
	50,001 - 100,000 THB	67	20.7%
	Above 100,000 THB	16	5.0%
6. On average, how often do you shop online?	Almost everyday (6 - 7 times per week)	17	5.3%
	Several times a week (2 - 5 times per week)	87	26.9%
	A few times a month (2 - 4 times per week)	145	44.9%
	Rarely (Less than once per month)	74	22.9%
7. Which platform do you normally use when shopping online?	Lazada	37	11.5%
	Shopee	190	58.8%
	Both Lazada and Shopee	96	29.7%

Fig 1. Demographics Table

To determine if a survey is usable, there exists several metrics to validate the data. One such metric is called the Cronbach's alpha. In this survey, the data will be cleaned and tested using Cronbach's alpha testing and Confirmatory Factor Analysis to determine whether removal of questions will increase the accuracy or reliability.

The data collected from the questionnaire is dirty in its original form and requires a clean up step to ensure that the data does not contain invalid inputs from users such as, straight lining or missing answers. 'Straight-lining' is the term used to describe selection of the same option all the way through in a multiple choice based question and is one type of invalid data in questionnaires. To potentially detect this, questions can flip positive and negative implications. One other method that was applied to the data set was z-score cleaning. It is the step to test the standard deviation of items and removal of outliers based on data plots that lie above or below near 2 standard deviation differences. For this research, it was determined that it is best to remove straight-lining, invalid replies, and z-score outliers before continuing with our testing for reliability and consistency.

The basis for our consistency testing is Cronbach's alpha testing. It can be defined as how an item closely relates to the group as a whole. A minimum score of 0.7 is required to be considered an acceptable reliability. In the testing, each question section is separated before scoring to determine each section's reliability before combination. The testing steps also include steps to remove certain questions before testing to discover weighted values of each question.

Data Reliability and Validation

To ensure that the questionnaire accurately reflects each conceptual construct such as motivation, engagement, loyalty, and gamification elements, reliability and validity testing were conducted.

This step is important because the quality of subsequent analysis depends on whether the questionnaire items are both consistent (reliability) and representative (validity).

1. Reliability Testing (Cronbach's Alpha)

Cronbach's Alpha (α) measures the internal consistency of a group of survey items, showing how closely related they are in measuring the same concept.

It is a widely accepted method for evaluating questionnaire reliability in social science and behavioral research.

Interpretation standards for Cronbach's Alpha:

- $\alpha \geq 0.9$ = Excellent
- $0.8 \leq \alpha < 0.9$ = Good
- $0.7 \leq \alpha < 0.8$ = Acceptable
- $0.6 \leq \alpha < 0.7$ = Questionable
- $0.5 \leq \alpha < 0.6$ = Poor
- $\alpha < 0.5$ = Unacceptable (requires revision)

Each construct was analyzed separately in SPSS to determine its reliability before combining them for structural analysis. It should be noted that the scale for interpretation is highly dependent on the data that is being analyzed. For this research, we have determined that this scale is at an acceptable level.

1.1 Intrinsic Motivation

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
ITM1	14.73	7.155	.528	.755
ITM2	14.40	7.626	.506	.759
ITM3	14.26	7.291	.632	.717
ITM4	14.29	7.994	.544	.748
ITM5	14.55	7.416	.595	.729

Reliability Statistics

Cronbach's Alpha	N of Items
.782	5

Interpretation:

An alpha of 0.782 indicates acceptable reliability. This means the five items designed to measure intrinsic motivation such as enjoyment, satisfaction, or curiosity when using Shopee or Lazada are consistent with one another.

Respondents likely answered these questions in a coherent pattern, implying they shared a common understanding of internal enjoyment or satisfaction.

Seeing that the value is only slightly above 0.7, there are points of improvements that could raise reliability further. By looking at the value, this suggests that while intrinsic motivation was consistently perceived, it still includes subtle individual differences in interpretation.

1.2 Extrinsic Motivation

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
EXM1	10.50	4.698	.366	.547
EXM2	10.20	4.169	.427	.499
EXM3	11.00	3.866	.484	.450
EXM4	11.23	4.415	.276	.621

Reliability Statistics

Cronbach's Alpha	N of Items
.603	4

Interpretation:

This construct produced an alpha of 0.603, which is below the acceptable threshold.

Such a result indicates that the items measuring external rewards such as discounts, cashbacks, or promotions did not correlate strongly enough to represent a single dimension.

Closer inspection of the item-total statistics shows that EXM4 had a corrected item-total correlation of 0.276, the lowest among all items. This suggests that respondents answered that question differently, possibly due to the polarity of the replies collected.

This can be interpreted as participants' perception of external rewards varying widely. Some may respond strongly to discounts, while others may value point accumulation or voucher redemption differently.

1.3 Engagement

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
ENG1	12.63	10.164	.455	.772
ENG2	13.11	8.877	.624	.717
ENG3	12.94	9.195	.636	.716
ENG4	13.41	8.473	.610	.722
ENG5	13.05	9.762	.467	.770

Reliability Statistics

Cronbach's Alpha	N of Items
.781	5

Interpretation:

With α =0.781, this construct demonstrates good reliability.

Respondents answered engagement-related items such as frequency of interaction, time spent, and emotional involvement in a consistent manner.

This confirms that engagement in gamified platforms is a stable behavioral construct across participants.

This finding aligns with previous studies showing that users engaged with gamification mechanisms tend to exhibit consistent behaviors such as frequent app usage or continuous participation in point systems, reflecting stable user habits.

1.4 Loyalty

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
LOY1	10.89	4.646	.666	.699
LOY2	11.29	4.598	.543	.774
LOY3	10.58	5.257	.645	.719
LOY4	10.74	5.255	.562	.753

Reliability Statistics

Cronbach's Alpha	N of Items
.788	4

Interpretation:

The reliability score of 0.788 suggests acceptable internal consistency among loyalty-related questions such as platform preference and recommendation intent.

These items collectively represent the participants' long-term attachment and brand commitment toward Shopee or Lazada.

This supports the theoretical link that loyalty arises when engagement and motivation reinforce satisfaction, leading to repetitive and sustained use.

1.5 Transaction-Based Gamification

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
T1	14.11	11.463	.433	.890
T2	14.44	9.309	.735	.820
T3	14.30	9.018	.803	.801
T4	14.23	10.061	.695	.831
T5	14.28	9.691	.760	.814

Reliability Statistics

Cronbach's Alpha	N of Items
.863	5

Interpretation:

An alpha value of 0.863 shows stronger reliability than most questions.

Participants responded to discount, voucher, and reward-point items with high consistency.

Such features are extrinsic and reward-oriented but are clearly recognized by users as belonging to the same category of gamification.

This consistency indicates that Shopee and Lazada users commonly perceive monetary rewards as a unified motivational system.

1.6 Achievement-Based Gamification

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
A1	12.31	13.899	.408	.919
A2	12.93	10.976	.781	.838
A3	12.79	11.053	.799	.834
A4	12.63	11.338	.765	.843
A5	12.65	11.190	.842	.825

Reliability Statistics

Cronbach's Alpha	N of Items
.881	5

Interpretation:

The reliability coefficient of 0.881 represents excellent consistency.

Respondents viewed progress bars, tier upgrades, and challenge systems as part of a coherent achievement mechanism.

The high alpha shows that the questionnaire effectively captured the goal-oriented nature of achievement-based gamification, which tends to generate long-term engagement through clear progress feedback.

1.7 Social Influence-Based Gamification

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
S1	14.23	13.467	.691	.928
S2	14.54	11.814	.819	.905
S3	14.31	11.955	.852	.898
S4	14.31	12.855	.806	.907
S5	14.37	12.359	.858	.897

Reliability Statistics

Cronbach's Alpha	N of Items
.925	5

Interpretation:

This section achieved the highest reliability among all constructs.

A 0.925 alpha indicates that items such as leaderboards, referrals, and review systems are extremely consistent in representing social gamification elements.

It confirms that participants interpret peer comparison and social validation features in the same way, highlighting how social interaction strongly reinforces motivation and community belonging in e-commerce.

2. Validity Testing (Confirmatory Factor Analysis)

Confirmatory Factor Analysis (CFA) is an analysis method that displays the relation between items in a dataset. CFA was conducted using SPSS AMOS and CFA provides evidence of construct validity, which determines whether the questionnaire truly measures what it is intended to measure. In this process, the relationships between each question and its corresponding construct are represented by standardized factor loadings ranging from 0 to 1. A higher loading value indicates that the item strongly contributes to the underlying factor, while a lower value suggests weak representation or potential misinterpretation by respondents. Typically, a factor loading above 0.50 is considered acceptable, indicating that the item meaningfully contributes to the construct. Items with loadings below the 0.50 threshold are usually removed from analysis, as they demonstrate low communality and are not well represented in the dataset. In this study, such low-loading items are highlighted in red to clearly indicate which questions may require revision or exclusion. In addition, included below is also a CFA diagram.

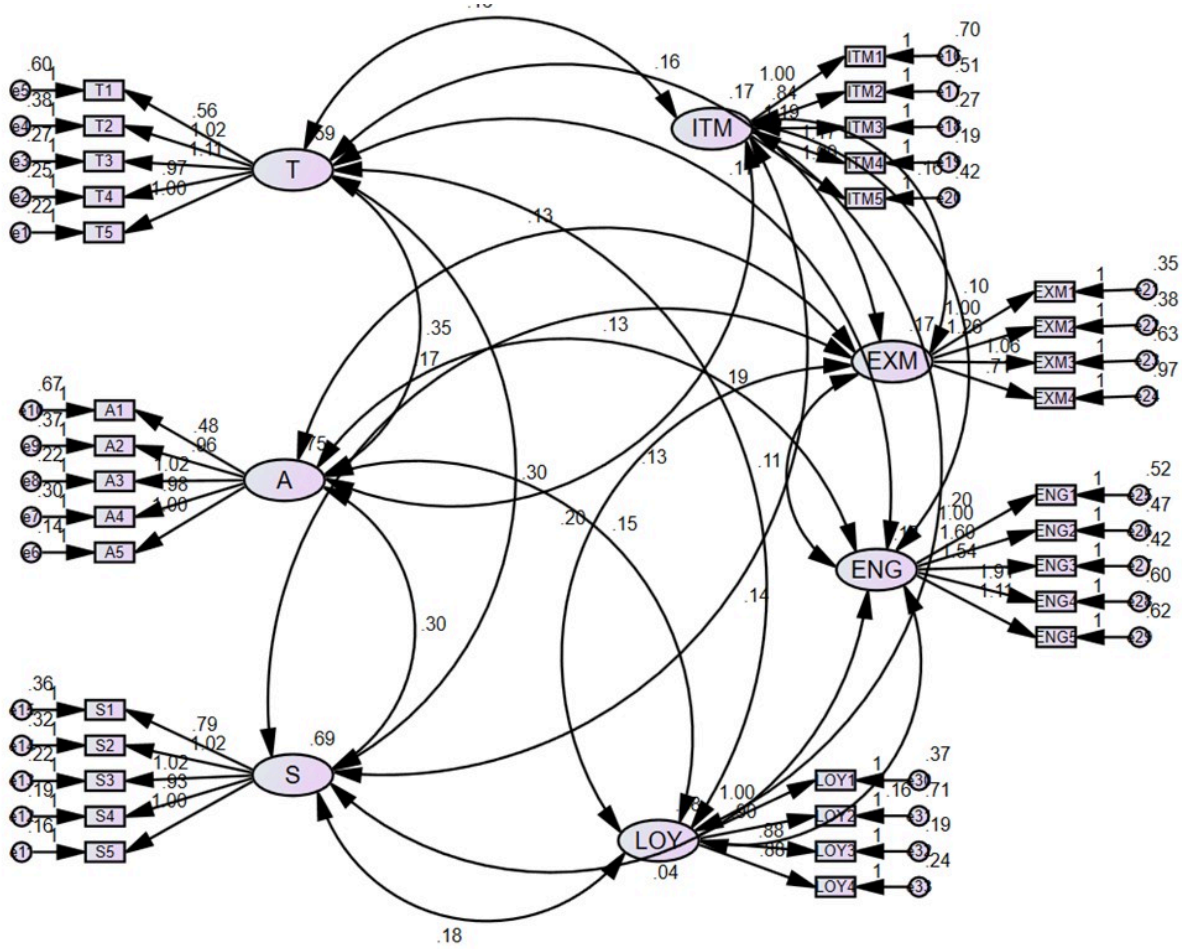


Fig 2. CFA Diagram

2.1 Transaction-Based Gamification

T5-T	.856
T4-T	.773
T3-T	.873
T2-T	.782
T1-T	.457

Analysis:

Excluding T1, all items display strong loadings above 0.79, meaning they are well aligned with the transaction-based gamification construct.

T1's lower loading (0.457) implies that one particular question, regarding acknowledgement of transaction-based gamification, did not strongly correlate with the others. This potentially reflects that not all users perceive every promotional event such as flash sales and coin cashbacks as part of the same mechanism.

However, the overall construct remains valid since four of five items surpass the threshold.

2.2 Achievement based Gamification

A5-A	.920
A4-A	.832
A3-A	.868
A2-A	.824
A1-A	.432

Analysis:

The majority of achievement-based items show very high factor loadings above 0.8, confirming excellent construct validity.

A1 = 0.432 is below 0.5 and represents an item that measures the acknowledgement of Achievement based Gamification. Removing this question could further improve model fit.

The strong loading of A5 (0.920) emphasizes that tier progression and challenge completion are the clearest indicators of user achievement perception.

2.3 Social influence based Gamification

S5-S	.915
S4-S	.859
S3-S	.892
S2-S	.843
S1-S	.708

Analysis:

All items demonstrate excellent convergent validity, ranging from 0.708 to 0.915.

This suggests that participants consistently relate these features to social validation and peer interaction.

The extremely high loading of S5 (0.915) may represent the item concerning top reviewer or community ranking, which aligns closely with the psychological principle of social recognition.

2.4 Intrinsic Motivations

ITM1-ITM	.592
ITM2-ITM	.571
ITM3-ITM	.741
ITM4-ITM	.672
ITM5-ITM	.680

Analysis:

All intrinsic motivation items have moderate to strong factor loadings between 0.571 and 0.730.

These results indicate potentially acceptable convergent validity, pointing towards intrinsic motivation items consistently measuring internal satisfaction and enjoyment when engaging with gamified e-commerce features.

The higher loadings of ITM3 and ITM4 show that users more strongly associate intrinsic motivation with feelings of personal satisfaction from the user experience.

The slightly lower loadings of ITM1 and ITM2 may be due to lacking competition within the application market within the area as the items are regarding the users choice in the usage of the platform.

2.5 Extrinsic Motivations

EXM1-EXM	.568
EXM2-EXM	.649
EXM3-EXM	.598
EXM4-EXM	.285

Analysis:

Factor loadings for Extrinsic Motivation range from 0.285 to 0.649.

While EXM1 through EXM3 are within acceptable limits, EXM4 is below 0.4, indicating weak representation of the construct.

This suggests that respondents did not perceive EXM4 as closely related to reward-driven behavior.

Such a pattern aligns with the earlier reliability test, confirming that some extrinsic items require removal or change to capture external motivation effectively.

Removing EXM4 could improve the model's construct validity.

2.6 Engagement

ENG1-ENG	.610
ENG2-ENG	.680
ENG3-ENG	.722
ENG4-ENG	.670
ENG5-ENG	.584

Analysis:

All items show factor loadings above 0.5, indicating good convergent validity.

ENG2, ENG3, and ENG4 have particularly strong correlations, suggesting that emotional and behavioral involvement are central to user engagement in e-commerce gamification.

The slightly lower value ENG5 may indicate that the item measures a different aspect due to the subject of the question.

2.7 Loyalty

LOY1-LOY	.674
LOY2-LOY	.549
LOY3-LOY	.788
LOY4-LOY	.773

Analysis:

All loyalty items exceed the 0.5 threshold, indicating strong construct validity.

The highest loadings are LOY3 and LOY4, showing that behavioral loyalty such as repeat purchases and preference continuity are highly representative of this construct.

The relatively lower value for LOY2 suggests that attitudinal loyalty (e.g., recommending to others) may be perceived differently from behavioral commitment, but still contributes meaningfully to overall loyalty.

An adjustment in the data is made to better fit the model. The data set below follows all prior steps, but this dataset includes the usage of z-score. The resulting Cronbach's Alpha results will display an overall better fit to the objective. Thus the change in the dataset. The new dataset is reflected in the section below

1.1 Intrinsic Motivation

Reliability Statistics

Cronbach's Alpha	N of Items
.698	5

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
ITM1	15.14	4.440	.412	.675
ITM2	14.83	4.935	.389	.676
ITM3	14.67	4.903	.480	.640
ITM4	14.71	4.961	.534	.624
ITM5	14.98	4.639	.494	.631

Interpretation:

An alpha of 0.698 indicates acceptable reliability. This means the five items designed to measure intrinsic motivation such as enjoyment, satisfaction, or curiosity when using Shopee or Lazada are consistent with one another.

Respondents likely answered these questions in a coherent pattern, implying they shared a common understanding of internal enjoyment or satisfaction.

Seeing that the value is only slightly below 0.7, there are points of improvements that could raise reliability further. By looking at the value, this suggests that while intrinsic motivation was consistently perceived, it still includes subtle individual differences in interpretation.

1.2 Extrinsic Motivation

Reliability Statistics

Cronbach's Alpha	N of Items
.565	4

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
EXM1	10.75	3.877	.324	.515
EXM2	10.43	3.396	.423	.436
EXM3	11.21	3.207	.397	.450
EXM4	11.52	3.230	.277	.569

Interpretation:

This construct produced an alpha of 0.565, which is below the acceptable threshold.

Such a result indicates that the items measuring external rewards such as discounts, cashbacks, or promotions did not correlate strongly enough to represent a single dimension.

Closer inspection of the item-total statistics shows that EXM4 had a corrected item-total correlation of 0.277, the lowest among all items. This suggests that respondents answered that question differently, possibly due to the polarity of the replies collected.

This can be interpreted as participants' perception of external rewards varying widely. Some may respond strongly to discounts, while others may value point accumulation or voucher redemption differently.

1.3 Engagement

Reliability Statistics

Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of Items
.764	.761	5

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
ENG1	12.91	9.138	.397	.215
ENG2	13.41	7.577	.625	.418
ENG3	13.26	7.818	.615	.401
ENG4	13.73	6.892	.624	.465
ENG5	13.31	8.717	.420	.191

Interpretation:

With $\alpha = 0.764$, this construct demonstrates good reliability.

Respondents answered engagement-related items such as frequency of interaction, time spent, and emotional involvement in a consistent manner.

This confirms that engagement in gamified platforms is a stable behavioral construct across participants.

This finding aligns with previous studies showing that users engaged with gamification mechanisms tend to exhibit consistent behaviors such as frequent app usage or continuous participation in point systems, reflecting stable user habits.

1.4 Loyalty

Reliability Statistics

Cronbach's Alpha	N of Items
.782	4

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
LOY1	11.07	3.695	.681	.677
LOY2	11.46	3.610	.548	.766
LOY3	10.77	4.397	.628	.718
LOY4	10.88	4.506	.542	.752

Interpretation:

The reliability score of 0.782 suggests acceptable internal consistency among loyalty-related questions such as platform preference and recommendation intent.

These items collectively represent the participants' long-term attachment and brand commitment toward Shopee or Lazada.

This supports the theoretical link that loyalty arises when engagement and motivation reinforce satisfaction, leading to repetitive and sustained use.

1.5 Transaction-Based Gamification

Reliability Statistics

Cronbach's Alpha	N of Items
.873	5

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
T1	14.24	11.178	.454	.900
T2	14.53	9.059	.760	.831
T3	14.39	8.858	.804	.819
T4	14.35	9.726	.737	.837
T5	14.37	9.602	.761	.832

Interpretation:

An alpha value of 0.873 shows stronger reliability than most questions.

Participants responded to discount, voucher, and reward-point items with high consistency.

Such features are extrinsic and reward-oriented but are clearly recognized by users as belonging to the same category of gamification.

This consistency indicates that Shopee and Lazada users commonly perceive monetary rewards as a unified motivational system.

1.6 Achievement-Based Gamification

Reliability Statistics

Cronbach's Alpha	N of Items
.885	5

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
A1	12.47	12.903	.431	.919
A2	13.05	10.341	.774	.847
A3	12.90	10.368	.812	.838
A4	12.79	10.493	.771	.848
A5	12.78	10.508	.842	.833

Interpretation:

The reliability coefficient of 0.885 represents excellent consistency.

Respondents viewed progress bars, tier upgrades, and challenge systems as part of a coherent achievement mechanism.

The high alpha shows that the questionnaire effectively captured the goal-oriented nature of achievement-based gamification, which tends to generate long-term engagement through clear progress feedback.

1.7 Social Influence-Based Gamification

Reliability Statistics

Cronbach's Alpha	N of Items
.925	5

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
S1	14.40	11.764	.716	.924
S2	14.68	10.530	.806	.909
S3	14.45	10.691	.838	.901
S4	14.46	11.229	.825	.905
S5	14.53	10.930	.845	.900

Interpretation:

This section achieved the highest reliability among all constructs.

A 0.925 alpha indicates that items such as leaderboards, referrals, and review systems are extremely consistent in representing social gamification elements.

It confirms that participants interpret peer comparison and social validation features in the same way, highlighting how social interaction strongly reinforces motivation and community belonging in e-commerce.

2. Validity Testing (Confirmatory Factor Analysis)

Confirmatory Factor Analysis (CFA) is an analysis method that displays the relation between items in a dataset. CFA was conducted using SPSS AMOS and CFA provides evidence of construct validity, which determines whether the questionnaire truly measures what it is intended to measure. In this process, the relationships between each question and its corresponding construct are represented by standardized factor loadings ranging from 0 to 1. A higher loading value indicates that the item strongly contributes to the underlying factor, while a lower value suggests weak representation or potential misinterpretation by respondents. Typically, a factor loading above 0.50 is considered acceptable, indicating that the item meaningfully contributes to the construct. Items with loadings below the 0.50 threshold are usually removed from analysis, as they demonstrate low communality and are not well represented in the dataset. In this study, such low-loading items are highlighted in red to clearly indicate which questions may require revision or exclusion

2.1 Transaction-Based Gamification

T5-T	.855
T4-T	.834
T3-T	.853
T2-T	.786
T1-T	.483

Analysis:

Excluding T1, all items display strong loadings above 0.79, meaning they are well aligned with the transaction-based gamification construct.

T1's lower loading (0.457) implies that one particular question, regarding acknowledgement of transaction-based gamification, did not strongly correlate with the others. This potentially reflects that not all users perceive every promotional event such as flash sales and coin cashbacks as part of the same mechanism.

However, the overall construct remains valid since four of five items surpass the threshold.

2.2 Achievement based Gamification

A5-A	.916
A4-A	.841
A3-A	.884
A2-A	.808
A1-A	.451

Analysis:

The majority of achievement-based items show very high factor loadings above 0.8, confirming excellent construct validity.

A1 = 0.432 is below 0.5 and represents an item that measures the acknowledgement of Achievement based Gamification. Removing this question could further improve model fit.

The strong loading of A5 (0.920) emphasizes that tier progression and challenge completion are the clearest indicators of user achievement perception.

2.3 Social influence based Gamification

S5-S	.902
S4-S	.871
S3-S	.876
S2-S	.831
S1-S	.739

Analysis:

All items demonstrate excellent convergent validity, ranging from 0.708 to 0.915.

This suggests that participants consistently relate these features to social validation and peer interaction.

The extremely high loading of S5 (0.915) may represent the item concerning top reviewer or community ranking, which aligns closely with the psychological principle of social recognition.

2.4 Intrinsic Motivations

ITM1-ITM	.439
ITM2-ITM	.434
ITM3-ITM	.683
ITM4-ITM	.738
ITM5-ITM	.570

Analysis:

All intrinsic motivation items have moderate to strong factor loadings between 0.571 and 0.730.

These results indicate potentially acceptable convergent validity, pointing towards intrinsic motivation items consistently measuring internal satisfaction and enjoyment when engaging with gamified e-commerce features.

The higher loadings of ITM3 and ITM4 show that users more strongly associate intrinsic motivation with feelings of personal satisfaction from the user experience.

The slightly lower loadings of ITM1 and ITM2 may be due to lacking competition within the application market within the area as the items are regarding the users choice in the usage of the platform.

2.5 Extrinsic Motivations

EXM1-EXM	.580
EXM2-EXM	.649
EXM3-EXM	.488
EXM4-EXM	.287

Analysis:

Factor loadings for Extrinsic Motivation range from 0.285 to 0.649.

While EXM1 through EXM3 are within acceptable limits, EXM4 is below 0.4, indicating weak representation of the construct.

This suggests that respondents did not perceive EXM4 as closely related to reward-driven behavior.

Such a pattern aligns with the earlier reliability test, confirming that some extrinsic items require removal or change to capture external motivation effectively.

Removing EXM4 could improve the model's construct validity.

2.6 Engagement

ENG1-ENG	.512
ENG2-ENG	.708
ENG3-ENG	.715
ENG4-ENG	.726
ENG5-ENG	.516

Analysis:

All items show factor loadings above 0.5, indicating good convergent validity.

ENG2, ENG3, and ENG4 have particularly strong correlations, suggesting that emotional and behavioral involvement are central to user engagement in e-commerce gamification.

The slightly lower value ENG5 may indicate that the item measures a different aspect due to the subject of the question.

2.7 Loyalty

LOY1-LOY	.709
LOY2-LOY	.548
LOY3-LOY	.776
LOY4-LOY	.746

Analysis:

All loyalty items exceed the 0.5 threshold, indicating strong construct validity.

The highest loadings are LOY3 and LOY4, showing that behavioral loyalty such as repeat purchases and preference continuity are highly representative of this construct.

The relatively lower value for LOY2 suggests that attitudinal loyalty (e.g., recommending to others) may be perceived differently from behavioral commitment, but still contributes meaningfully to overall loyalty.

Based on the factors of low-loading items that were mentioned above, the figures are removed resulting in this table below:

Transaction-Based Gamification

T5-T	.855
T4-T	.834
T3-T	.853
T2-T	.786

Achievement-based Gamification

A5-A	.916
A4-A	.841
A3-A	.884
A2-A	.808

Social influence based Gamification

S5-S	.902
S4-S	.871
S3-S	.876
S2-S	.831
S1-S	.739

Intrinsic Motivations

ITM3-ITM	.683
ITM4-ITM	.738
ITM5-ITM	.570

External Motivations

EXM1-EXM	.580
EXM2-EXM	.649

Engagement

ENG1-ENG	.512
ENG2-ENG	.708
ENG3-ENG	.715
ENG4-ENG	.726
ENG5-ENG	.516

Loyalty

LOY1-LOY	.709
LOY2-LOY	.548
LOY3-LOY	.776
LOY4-LOY	.746

This result poses an issue in a future step (SEM: Structural Equation Modelling) as External motivation only has 2 data plots left. This issue can be resolved by lowering the cutoff down to 0.3 rather than 0.5 to ensure that External motivation has more than 2 data plots to use in SEM. Below are the tables which are affected and their changes:

Intrinsic Motivation

ITM1-ITM	.439
ITM2-ITM	.434
ITM3-ITM	.683
ITM4-ITM	.738
ITM5-ITM	.570

Achievement-based Gamification

A5-A	.916
A4-A	.841
A3-A	.884
A2-A	.808
A1-A	.451

Transaction-Based Gamification

T5-T	.855
T4-T	.834
T3-T	.853
T2-T	.786
T1-T	.483

External Motivations

EXM1-EXM	.580
EXM2-EXM	.649
EXM3-EXM	.488

3. Item relation (Convergent Validity)

Convergent Validity is a test to display relation between items which should, in theory, be related. In this section, we will produce 2 results, composite reliability and average reliability. The inputs for the calculation will be results of the CFA in section 2.

Composite Reliability:

(a)

$$CR = \frac{(\sum \lambda)^2}{\left[(\sum \lambda)^2 + \sum (1 - \lambda^2)\right]}$$

Fig. 3 Composite Reliability Equation

Variable	⋏	⋏2	1-⋏2
T (Transaction-based gamification)	0.79	0.6241	0.3759
	0.857	0.73445	0.26555
	0.829	0.68724	0.31276
	0.848	0.7191	0.2809
A (Achievement-based gamification)	0.811	0.65772	0.34228
	0.884	0.78146	0.21854
	0.839	0.70392	0.29608
	0.907	0.82265	0.17735
S (Social influence-based gamification)	0.739	0.54612	0.45388
	0.826	0.68228	0.31772
	0.879	0.77264	0.22736
	0.863	0.74477	0.25523
	0.901	0.8118	0.1882
ITM (Intrinsic Motivation)	0.703	0.49421	0.50579
	0.73	0.5329	0.4671
	0.591	0.34928	0.65072
	0.566	0.32036	0.67964
	0.656	0.43034	0.56966
ETM (Extrinsic Motivation)	0.521	0.27144	0.72856
	0.554	0.30692	0.69308
	0.719	0.51696	0.48304
ENG (Engagement)	0.719	0.51696	0.48304
	0.713	0.50837	0.49163
	0.511	0.26112	0.73888
	0.703	0.49421	0.50579
	0.543	0.29485	0.70515
LOY (Loyalty)	0.779	0.60684	0.39316
	0.759	0.57608	0.42392
	0.79	0.6241	0.3759
	0.857	0.73445	0.26555

Table 1. Composite Reliability

Average Reliability:

(b)

$$AVE = \frac{\sum \lambda^2}{n}$$

Fig. 4 Average Reliability Equation

Using the same values from CR will result in: AR = 0.69122 with a square root average of 0.8314.

From these formulas, when applied to all items will result in the table below:

Variable	CR	AR	SQR AR
T (Transaction-based gamification)	0.89945	0.69122	0.83140
A (Achievement-based gamification)	0.91967	0.74144	0.86107
S (Social influence-based gamification)	0.92468	0.71152	0.84352
ITM (Intrinsic Motivation)	0.71616	0.45880	0.67735
EXM (Extrinsic Motivation)	0.60568	0.34071	0.58370
ENG (Engagement)	0.78162	0.42207	0.64967
LOY (Loyalty)	0.79261	0.49300	0.70214

Table 2. Convergent Validity

Following the rule of thumb for Composite Reliability, we can note that only EXM is lower than 0.7 which is considered to be 'Good'. To confirm that these values are valid and usable, the values are tested using **Fornell–Larcker criterion**.

The **Fornell–Larcker criterion** is a discriminant validity test that identifies the degree to which each construct differs from each other. To perform testing, the values are computed in SPSS AMOS using a built-in function, in which the goal for testing is to determine if all constructs are unique from each other by comparing SQR AR values. If all downwards diagonal values, **self comparison values**, are the highest values. Then, it can be determined that all constructs are unique.

Construct	Mean	SD	ITM	EXM	ENG	LOY	T	A	S
ITM	3.6111	0.66659	*0.67735*						
EXM	3.5782	0.64861	.582**	*0.5837*					
ENG	3.257	0.74215	.512**	.522**	*0.78162*				
LOY	3.6246	0.71594	.544**	.510**	.573**	*0.79261*			
T	3.5678	0.77376	.395**	.365**	.347**	.362**	*0.8314*		
A	3.1659	0.84233	.348**	.308**	.411**	.346**	.496**	*0.86107*	
S	3.5882	0.87533	.352**	.351**	.201**	.283**	.459**	.419**	*0.84352*

Table 3. Construct Correlations Table

3. SEM-AMOS results

Structural Equation Modeling (SEM) is a varied set of techniques and methods used in statistical studies that allows the observation of direct and indirect relationships between complex variables. This allows researchers to determine significant relationships and model fits between variables.

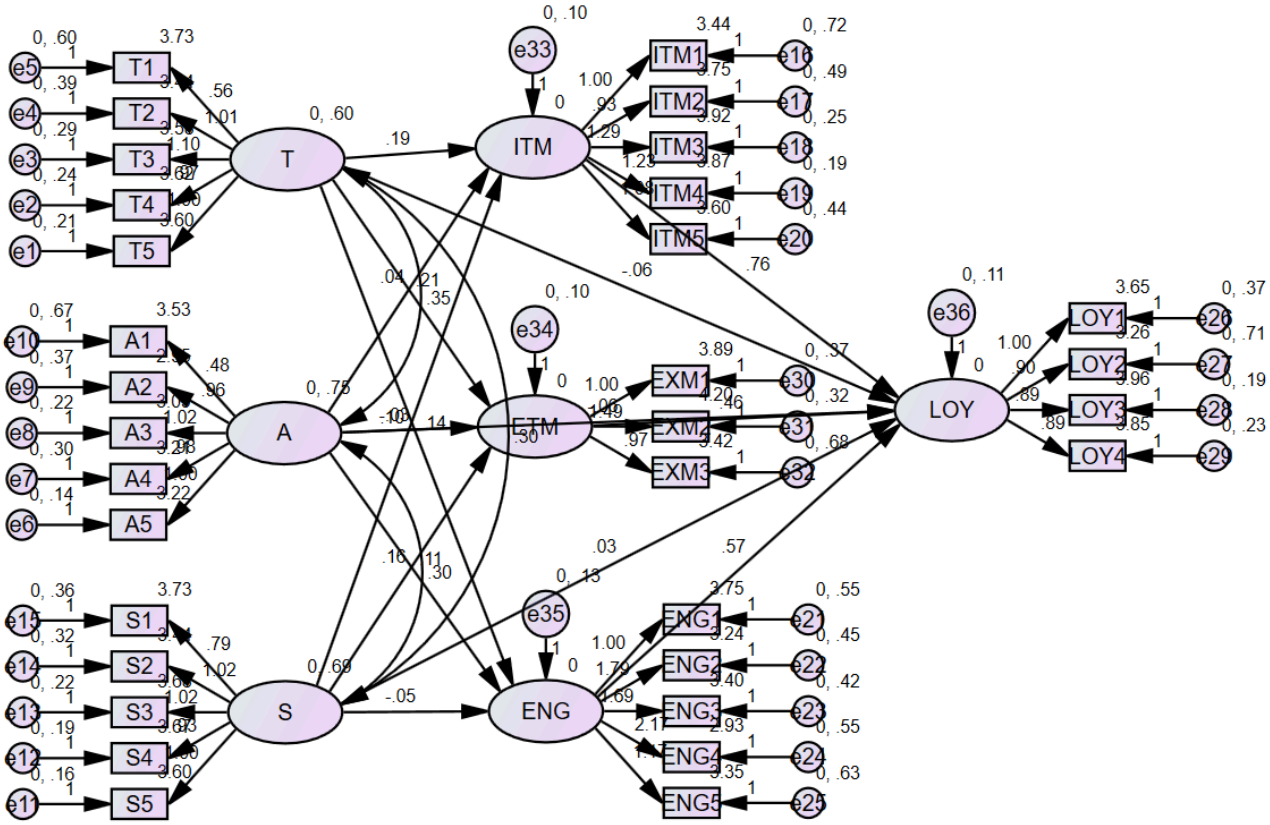


Fig. 5 SEM diagram displaying relationship between constructs

3.1 Direct effects

By utilizing tools in SPSS AMOS, the dataset is evaluated for its regression weights. In this computation, an estimate is first calculated by maximum likelihood estimate. After which, a standard error for that estimate is computed. With these two values a new value, critical ratio, is calculated by dividing the standard error by the estimate (Estimate/Standard Error). AMOS then converts the critical ratio into significance. This significance value indicates the significance of the relationship between items in the dataset.

Significance is indicated by the asterisk symbol in the column ‘P’. Three asterisk symbols indicate ‘Very highly significant’, two asterisk indicate ‘highly significant’, one asterisk indicates ‘significant’, and (ns) indicates ‘not significant’. These symbols are given based on the range given below.

*** ($p < 0.001$) → Very highly significant

** ($p < 0.01$) → Highly significant

* ($p < 0.05$) → Significant

ns ($p > 0.05$) → Not significant

	Estimates	**S.E.**	**C.R.**	**P**
T → ITM	.371	.049	3.871	***
T → EXM	.418	.051	4.188	***
T → ENG	.284	.045	3.162	.002**
A → ITM	.084	.035	1.093	.274 (ns)
A → EXM	-.062	.038	-.747	.455 (ns)
A → ENG	.234	.038	2.805	.005**
S → ITM	.220	.037	2.798	.005**
S → EXM	.336	.042	3.829	***
S → ENG	-.110	.036	-1.444	.149 (ns)
T → LOY	-.086	.067	-.936	.349 (ns)
A → LOY	-.091	.046	-1.294	.196 (ns)
S → LOY	.045	.051	.602	.547 (ns)
ITM → LOY	.531	.162	4.722	***
EXM → LOY	.318	.148	3.096	.002**
ENG → LOY	.391	.121	4.690	***

Fig. 6 Direct effects

The negative estimate values indicate a negative effect on the constructs relation. For example, Achievement-based gamification has a negative effect on extrinsic motivation. However, this effect is also non-significant as indicated by the significance level.

Based on the resulting table above, we can conclude that six relations do not pose a significant positive benefit to loyalty; the relations are as follows: Achievement-based Gamification→Intrinsic Motivator, Extrinsic Motivator, and Loyalty, Social Influence-based Gamification→ Engagement and Loyalty, and Transaction-based Gamification → Loyalty.

3.2 Indirect effects (Mediating effect)

Mediating effect refers to the relationship between an independent variable and a dependent variable through a mediating variable. In this step, AMOS will calculate the mediation effect from predictor variables to the final variable, in this case, Loyalty. This is calculated by using the estimates of the path from predictor to the mediator multiplied by the estimate of the path from the mediator to the Loyalty. For example, T → ITM → LOY will be calculated using (.371 ✕ .531 = 0.197). These values are then summed to create a total effect. Where this total effect is derived from both direct and indirect effects. Each of their values will also compute for significance (P) to determine the results of mediation with the focus on Indirect effects. Their determination of significance utilizes the same range as the Direct effects table.

*** ($p < 0.001$) → Very highly significant

** ($p < 0.01$) → Highly significant

* ($p < 0.05$) → Significant

ns ($p > 0.05$) → Not significant

Path	**Total Effects**	**Direct Effects**	**Indirect Effects**	**Results**
T → ITM →LOY T → EXM →LOY T → ENG →LOY	.259**	-.063	.322**	Have mediating effect
P - values	(0.010)	(0.280)	(0.010)	
A → ITM →LOY A → EXM →LOY A → ENG →LOY	.016	-.059	.075	No mediating effect
P - values	(0.772)	(0.184)	(0.239)	
S → ITM →LOY S → EXM →LOY S → ENG →LOY	.154*	.032	.123	No mediating effect
P - values	(0.028)	(0.634)	(0.056)	

Fig. 7 Indirect Effects

Following the same significance value evaluation as before, we can conclude that only Transaction-based Gamification has a mediating effect in an indirect manner.

3.3 Model Fit

By using the AMOS tool, it is also possible to extract Chi-square divided by degrees of freedom (CMIN/DF) which displays how well the model reproduces its data, or in other terms, how well the model represents the data. Typically, CMIN/DF is used for sample sizes between 75 and 200 with the interpretation as follows:

≤ 2.0 : Excellent fit

2.0 – 3.0 : Acceptable / good fit

3.0 – 5.0 : Moderate / marginal fit

> 5.0 : Poor fit

Table 4. CMIN/DF result from SPSS AMOS

Model	**CMIN/DF**
Default model	3.193
Saturated model	
Independence model	11.725

CMIN/DF represents the absolute fit of the model, whilst a comparative fit is also creatable using SPSS AMOS.

Comparative fit, also referred to as Baseline Comparison, displays a comparison between the model and a set baseline. This table is also created by SPSS AMOS automatically with the model. The interpretation of the values are shown below.

CFI (Comparative Fit Index) → ≥ 0.80 (good ≥ 0.90)

TLI (Tucker-Lewis Index) → ≥ 0.80 (good ≥ 0.90)

NFI (Normed Fit Index) → ≥ 0.80 (good ≥ 0.90)

RMSEA (Root Mean Square Error of Approximation) → ≤ 0.08 (good ≤ 0.05)

Table 5: Baseline Comparisons

Model	NFI Delta1	RFI rho1	IFI Delta2	TLI rho2	CFI
Default model	.755	.728	.818	.795	.816
Saturated model	1.000		1.000		1.000
Independence model	.000	.000	.000	.000	.000

Table 6: RMSEA (Root Mean Square Error of Approximation)

Model	RMSEA	LO 90	HI 90	PCLOSE
Default model	.086	.081	.091	.000
Independence model	.191	.187	.195	.000

The values for TLI, NFI, and RMSEA are at 0.795, 0.755, and 0.086. These values can be interpreted as reasonably well fitting with room for improvement. Displaying that the model is marginally well fitting. Given that the RMSEA value is slightly above the recommended cutoff indicates a moderate approximation error.

4. Discussions and Future Work

Reflecting upon the work that has been completed by the researchers, there are a few notable challenges. The most difficult of which being the sample of the dataset. The members that were in charge of conducting the questionnaire were unable to find the most diverse source of data. This result creates a skew in the population of the data. This change may lead to a differing result and, thus, conclusion. There are points in the analysis that require the usage of ‘rule of thumb’s that may change in the future and alter the resulting data creating a change in the final step and causing a different significance value.

Considering the hypotheses that were originally made, the research was based on the hypothesis that all elements will have a positive effect on loyalty. With the intermediate constructs being expected to display a positive mediating effect. From the resulting values, there were unexpected results with no significant indirect effects with the exception of the transaction-based gamification.

The model was tested for fitting and the results are on the lower end of the acceptable range, this can be interpreted as the model representing the hypothesis but further research and evaluation may be needed to decisively determine the results.

5. Conclusion

Based on the data that was collected and analyzed using multiple steps in SPSS AMOS, the final significance values indicate that Transaction-based Gamification, with a value of 0.002 or lower in direct effect and 0.01 in indirect effects, has the best overall effects on Loyalty. Thus, the results recommend creating a Transaction-based Gamification as the core mechanic of gamification with the support of the other methods to potentially improve likelihood of loyalty. Overall, the results can be considered good and acceptable. However, there may be benefits to increasing the dataset. There are points in the analysis that may also benefit from adjustment in thresholds.